Queue Theory: Part I

ORSA Study text

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Queuing Models – Waiting lines

QUEUE

Any system where jobs (or customers, users) arrive looking for service and depart once service is provided. A schematic of a typical queue is shown below:

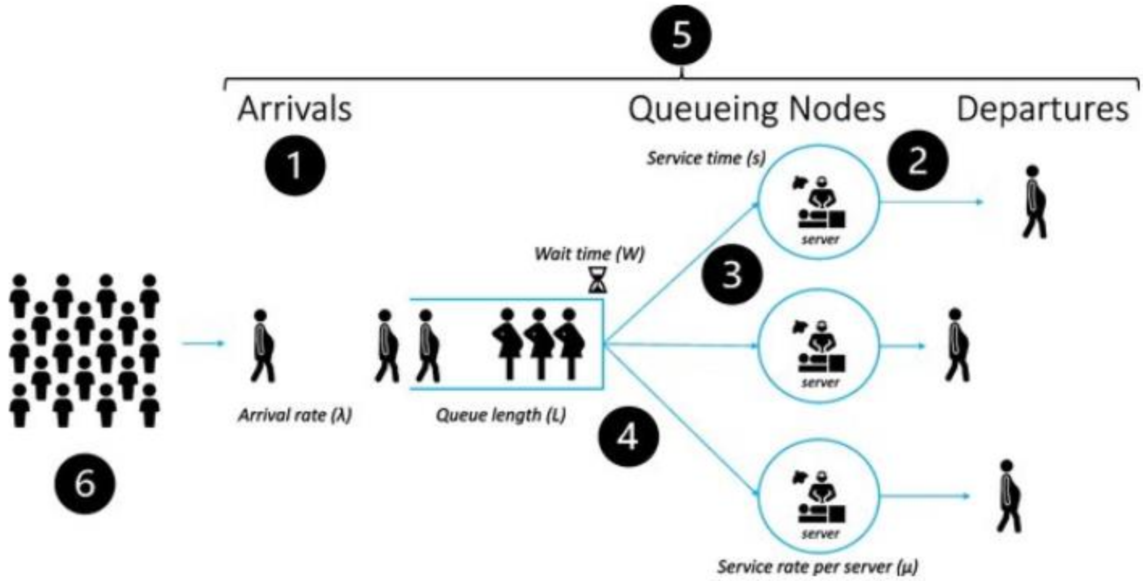

Figure 1 Source: Lim, G., Lim, A.J., Quinn, B. *et al.* Obstetric operating room staffing and operating efficiency using queueing theory. *BMC Health Serv Res* **23**, 1147 (2023). https://doi.org/10.1186/s12913-023-10143-0

Some Examples Where we Encounter Queuing

- Waiting to cash a check in the bank - as a matter of fact, any situation where we need to stand in line to obtain service from some server
- The telephone system - this provided the classical motivation for studying the behaviour of queues. A queuing situation is typically encountered when a number of customers try to call in order to gain access to one of a finite set of lines going out from the exchange
- Computer Networks - data packets/messages get forwarded from source to destination, possibly through one or more intermediate nodes. Queuing would arise at each such point where buffering and forwarding is done.
- Computer Systems - here the computing and other jobs in the system require service from one or more system resources giving rise to a queuing type model.
- Queuing theory may be used to determine the optimum number of:
- Toll booths for a bridge or toll road
- Doctors available for clinic calls.
- Repair persons servicing machines.
- Landing strips for aircraft.
- Docks for ships.
- Paramedic units available for emergency calls.
- Clerks for a spare-parts counter.

- Service Windows for a post office.

The above are only a few examples of the many different waiting-line situations encountered by managers.

Modelling Issues and Performance

Queuing theory or waiting line theory is primarily concerned with processes characterised by random arrivals (i.e., arrivals at random time intervals); the servicing of the customer is also a random process. If we assume there are costs associated with waiting in line, and if there are costs of adding more channels (i.e., adding more service facilities), we want to minimise the sum of the costs of waiting and the costs of providing service facilities. The computations will lead to such measures as the expected number of people in line, the expected waiting time of the arrivals, and the expected percentage utilisation of the service facilities. These measures can then be used in the cost computations to determine the number and capacity of service facilities that are desirable.

a) Most frequent modelling issues in the study of queues and queuing:

Arrival Process Description.
How does one describe the process of customer arrivals?

Buffer Size/lNumber of Waiting Positions.
Is there any provision to wait in the queue in case the server(s) are currently busy?

Service Discipline.
Are there any local rules governing the order in which service is provided and/or the type of service?

Service Time Distribution.
What is the length of service and what is its distribution if this is a random variable?

Priorities.
Are there any service priorities involved which decides between customers on who should be serviced first?

b) Most frequent performance issues in the study of queues and queuing:

Mean Service Parameters such as mean waiting times, mean number in queue etc..

Distribution of Service Parameters or their transforms - much harder to find usually than the means but would be nice to know them, if possible.

Server utilisation.

Number and distribution of customers leaving without service, if any.

Description of the customer departure process - usually very hard to find.

Economics of the Waiting Line Problem

Understanding waiting lines or queues and learning how to manage them is one of the most important areas in operations management. It is basic to creating schedules, job design, inventory levels, and so on. In our service economy we wait in line every day, from driving to work to checking out at the supermarket. We also encounter waiting lines at factories-jobs wait in lines to be worked on at different machines, and machines themselves wait their turn to be overhauled. In short, waiting lines are pervasive.

In this section we discuss the basic elements concerning economics and decision making of waiting line problems and provide standard steady-state formulas for solving them. These formulas, arrived at through queuing theory, enable economists and planners to analyse service requirements and establish service facilities appropriate to stated conditions. Queuing theory is broad enough to cover such dissimilar delays as those encountered by customers in a shopping mall or aircraft in a holding pattern awaiting landing slots.

The central problem in virtually every waiting line situation is a trade-off decision. The manager must weigh the added cost of providing more rapid service (more traffic lanes, additional landing strips, more checkout stands) against the inherent cost of waiting.

Frequently the cost trade-off decision is straightforward. For example, if we find that the total time our employees spend in line waiting to use a copying machine would otherwise be spent in productive activities, we could compare the cost of installing one additional machine to the value of employee time saved. The decision could then be reduced to dollar terms and the choice easily made.

On the other hand, suppose that our waiting line problem centres on demand for beds in a hospital. We can compute the cost of additional beds by summing the costs for building construction, additional equipment required, and increased maintenance. But what is on the other side of the scale? Here we are confronted with the problem of trying to place a dollar figure on a patient's need for a hospital bed that is unavailable. While we can estimate lost hospital income, what about the human cost arising from this lack of adequate hospital care?

Figure 2 shows the essential trade-off relationship under typical (steady-state) customer traffic conditions. Initially, with minimal service capacity, the waiting line cost is maximum. As service capacity is increased, there is a reduction in the number of customers in the line and in their waiting times, which decreases waiting line cost.

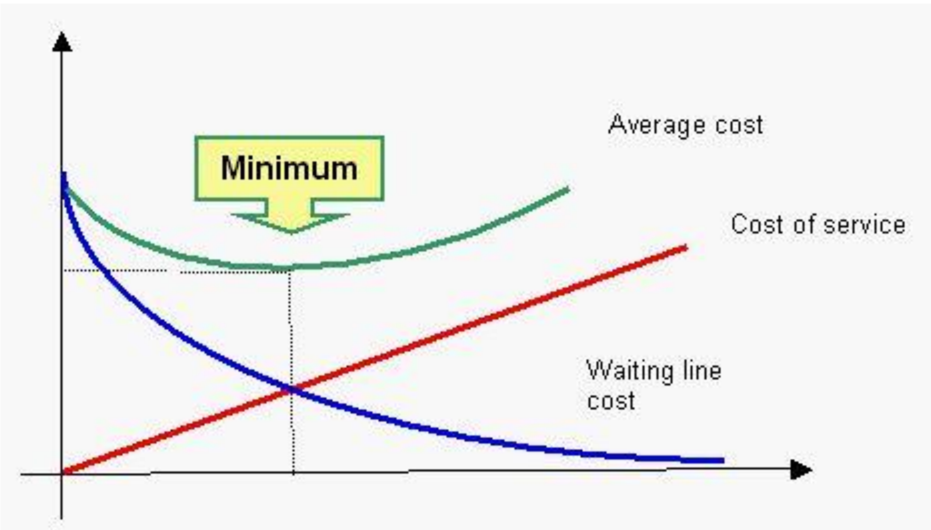

Figure 2 Source: Distance Learning Module for Management Science. Online. Available from: https://orms.pef.czu.cz/.

Finite and Infinite Population in Queues

Arrivals at a service system may be drawn from *a finite* or an *infinite* population. The distinction is important because the analyses are based on different premises and require different equations for their solution.

Finite Population: A finite population refers to the limited size customer pool that will use the service and, at times, form a line. The reason this finite classification is important is because when a customer leaves its position as a member of the population of users (by a machine breaking down and requiring service, for example), the size of the user group is therefore reduced by one, which reduces the probability of the next occurrence. Conversely, when a customer is serviced and returns to the user group, the population increases and the probability of a user requiring service also increases. This finite class of problems requires a separate set of formulas from that of the infinite population case. See an Example of a finite population.

Infinite Population: An infinite population is one large enough in relation to the service system so that the changes in the population size caused by subtractions or additions to the population (a customer needing service or a serviced customer returning to the population) does not significantly affect the system probabilities. See an Example of an infinite population .

Queues and Waiting Lines in Everyday Life

Queues or waiting lines are very common in everyday life. There are few individuals in modem society who have not had to wait in line for a bus, a taxi, a movie ticket, a grocery check out, a haircut, or registration material at the beginning of the school year. Most of us consider lines an unavoidable part of our civilised life, and we put up with them with more or less good humour. Occasionally, the size of a line we encounter discourages us, we abandon the project, and a sale is lost.

Management of Queues

Fortunately for the busy manager, reasonable queuing decisions can frequently be based on past experience or on the facts of the current situation. Thus, the management of a grocery chain knows approximately how many check out counters should be installed in a new store by looking at the experience of comparable stores. At any time of the day, the store manager can tell how many of the installed counters should be open by noting the lengths of the queues and adding personnel from other chores, or sending the present checkout personnel to other tasks. Also, historical records (for example of machine downtime) can indicate the amount of time that machines had to queue for repairs rather than relying upon the computations resulting from a mathematical model to determine the amount of downtime.

Although a number of problems encountered by an executive can be reasonably solved by the use of intuition or past experience, there will be many situations that are too complex for our intuition or where we desire a more accurate answer than we can expect to be supplied by intuition. In these situations, the problem can be approached by either simulation or a mathematical model procedure.

MATHEMATICAL MODELLING OF QUEUING

Background to the Solution

There are two approaches to mathematical analysis of a queue:
1. **Analytical model**. Behaviour of all elements in the queue is described by system of (differential) equations.
2. **Simulation model**. Behaviour and relations among elements of the queue is modelled by computer programme.

In this section we describe in detail two models: a) the single-channel and b) the multiple-channel, both with infinite source and patient customers. These two types of queues might be thought of as the mainstream queuing models - this stems partly from the ease understanding of queuing systems that they convey and partly from the ease with which system performance measures can be obtained.

However, the field of queuing contains a rich variety of models - all these models can be solved as a slight or complicated variations in the assumptions which underlie the two basic models above.

Other queuing models one can solve using special SW for queues or by means of simulations.

In both analytical and simulation model there are four major elements of any queuing situation which must be considered: arrivals, services, number of servers, queue discipline.

Arrivals

Customers come into the system for service. How these customers arrive in the system is important:

- They may come *singly* or in *batches*.
- They may come in *deterministic* (evenly spaced in time) or in a *random* pattern.
- They may come from *an infinite population* of possible (i.e., a very large), or they may come from a *finite set*.

Distribution of Arrivals

When describing a waiting system, we need to define the manner in which customers or the waiting units are arranged for service.

Waiting line formulas generally require an arrival rate, or the number of units per period (such as 10 units per hour). The time between arrivals is the interarrival time (such as an average of one every six minutes). A *constant* arrival distribution is periodic, with exactly the same time period between successive arrivals. In productive systems, about the only arrivals that truly approach a constant interarrival period are those that are subject to machine control. Much more common are *variable* (random) arrival distributions.

In observing arrivals at a service facility, we can look at them from two viewpoints: We can analyse the time between successive arrivals to see if the times follow some statistical distribution. Usually, we assume that the time between arrivals is exponentially distributed. We can set some time length (t) and try to determine how many arrivals might enter the system within t. We will typically assume that the number of arrivals per time unit is Poisson distributed. See Exponential distribution.

Other Arrival Characteristics – Patterns, Size, Patience

Other arrival characteristics include arrival patterns, size of arrival units, and degree of patience:

Arrival patterns: The arrivals at a system are far more *controllable* than is generally recognized. Barbers may decrease their Saturday arrival rate (and supposedly shift it to other days of the week) by charging an extra $1 for adult haircuts or charging adult prices for children's haircuts. Department stores run sales during the off-season or hold one day-only sales in part for purposes of control. Airlines offer excursion and off-season rates for similar reasons. The simplest of all arrival-control devices is the posting of business hours.

Some service demands are clearly *uncontrollable*, such as emergency medical demands on a city's hospital facilities. But even in these situations, arrivals at emergency rooms in specific hospitals are controllable to some extent by, say, keeping ambulance drivers in the service region informed of the status of their respective host hospitals.

Size of arrival units: A single arrival may be thought of as one unit. (A unit is the smallest number handled.) A single arrival on the floor of the New York Stock Exchange is 100 shares of stock; a single arrival at an egg-processing plant might be a dozen eggs or a flat of 2t/2 dozen; or a single person at a restaurant.

A batch arrival: It is some multiple of the unit, as a block of 1,000 shares on the NYSE, a case of eggs at the processing plant, or a party of five at a restaurant.

Degree of patience: A patient arrival is one who waits as long as necessary until the service facility is ready to serve him or her. (Even if arrivals grumble and behave impatiently, the fact that they wait is sufficient to label them as patient arrivals for purposes of waiting line theory.)

There are two classes of *impatient* arrivals. Members of the first class at TGV, survey both the service facility and the length of the line, and then decide to leave. Those in the second class arrive, view the situation, join the waiting line, and then, after some period of time, depart. The behaviour of the first type is termed *balking*, while the second is termed *reneging*.

Queue and Services

The queuing system consists primarily of the waiting lines and the available number of servers. Factors to consider here include the line length, number of lines, and queue discipline. Each customer must be serviced. How long it takes to complete a service is the second important element. It may take the same time for each customer, or service times may vary considerably in a random fashion.

There may be only *one server* or channel or *many servers*. A customer may be processed by only *one server* or *several in turn*. Servers may have *different service rates*.

Length: In a practical sense, an infinite line is very long in terms of the of the service system. Examples of infinite potential length are a line of vehicles backed up for miles at a bridge crossing and customers who must form a line around the block as they wait to purchase tickets at a theatre.

Gas stations, loading docks, and parking lots have *limited line capacity* caused by legal restrictions or physical space characteristics. This complicates the waiting line problem not only in service system utilization and waiting line computations but also in the shape of the actual arrival distribution. The arrival denied entry into the line because of lack of space may rejoin, the population for a later try or may seek service elsewhere. Either action makes an obvious difference in the finite population case.

Number of lines*: A single line* is one line only. The term *multiple lines* refers either to the single lines that form in front of two or more servers onto single lines that converge at some central redistribution point. The disadvantage of multiple lines in a busy facility is that arrivals often shift lines if several previous services have been of short duration or if those customers currently in other lines appear to require a short service time.

Service Time Distribution

Another important feature of the waiting structure is the time the customer or unit spends with the server once the service has started. Waiting line formulas generally specify *service rate*as the capacity of the server in number of units per time period (such as 12 completions per hour) and not as service time, which might average five minutes each. *A constant* service time rule states that each service takes exactly the same time. As in constant arrivals, this characteristic is generally limited to machine-controlled operations.

When service times are random, a good approximation of them can be given by the exponential distribution . When using the exponential distribution as an approximation of the service times, we will refer to μ, as the average number of units or customers that can be served per time period.

Line Structures

The flow of items to be serviced may go through a single line, multiple lines, or some mixtures of the two. The choice of format depends partly on the volume of customers served and partly on the restrictions imposed by sequential requirements governing the order in which service must be performed.

The simplest type of waiting line structure is single server, single phase, and straightforward formulas are available to solve the problem for standard distribution patterns of arrival and service. When the distributions are non standard, the problem is easily solved by computer simulation. A typical example of a single server, single phase situation is the one-person barbershop.

Queue Discipline

While customers wait for service, they are in a waiting line or queue. There may be only one line or separate ones for each server. There may be a *space limit* on the waiting line, and customers who arrive when the line is full may turn away called *balking.*

Queue discipline: A queue discipline is a priority rule or set of rules for determining the order of service to customers in a waiting line. The rules selected can have a dramatic effect on the system's overall performance. The number of customers in line, the average waiting time, the range of variability in waiting time, and the efficiency of the service facility are just a few of the factors affected by the choice of priority rules.

Probably the most common priority rule is *first come, first served* (FCFS). This rule states that customers in line are served on the basis of their chronological arrival; no other characteristics have any bearing on the selection process. This is popularly accepted as the fairest rule although in practice, it discriminates against the arrival requiring a short service time.

Reservations first, emergencies first, highest profit customer first, largest orders first, best customers first, longest waiting time in line, and *soonest promised date* are other examples of priority rules. There are two major practical problems in using any rule: One is ensuring that customers know and follow the rule. The other is ensuring that a system exists to enable the employees to manage the line (e.g., take-a-number systems).